

$$\Rightarrow f(x) = \frac{3}{8x} - \frac{x}{8} + \frac{1}{4} \quad \dots(\text{iii})$$

100. Option (a) is correct.

$$\begin{aligned} 8 \int_1^2 f(x) dx &= 8 \int_1^2 \left(\frac{3}{8x} - \frac{x}{8} + \frac{1}{4} \right) dx \\ &= \int_1^2 \left(\frac{3}{x} - x + 2 \right) dx \\ &= \left[3 \ln x - \frac{x^2}{2} + 2x \right]_1^2 \\ &= \left(3 \ln 2 - \frac{4}{2} + 4 \right) - \left(3 \ln 1 - \frac{1}{2} + 2 \right) \\ &= 3 \ln 2 + 2 - \frac{3}{2} \\ &= \frac{1}{2} + 3 \ln 2 \\ &= \frac{1}{2} \ln e + \ln 2^3 \\ &= \ln \sqrt{e} + \ln 8 \\ &= \ln(8\sqrt{e}) \end{aligned}$$

101. Option (c) is correct.

Number of ways of selecting two black balls from the bag = 5C_2

Number of ways of selecting two white balls from the bag = 4C_2

Number of ways of selecting both of the ball of same colour = ${}^5C_2 + {}^4C_2$

Now, the required probability is,

$$P[E] = \frac{{}^5C_2 + {}^4C_2}{{}^9C_2} = \frac{10+6}{36}$$

$$P[E] = \frac{16}{36} = \frac{4}{9}$$

102. Option (c) is correct.

Mean = 5 and Variance = 4

$$5^{23}P(X=3) = \lambda 4^\lambda \quad \dots(\text{i})$$

$$\therefore \text{Mean} = 5$$

$$\Rightarrow np = 5$$

$$\text{Variance} = 4$$

$$\Rightarrow npq = 4$$

$$\Rightarrow 5q = 4$$

$$\Rightarrow q = \frac{4}{5}$$

$$\text{Now, } p = 1 - q = 1 - \frac{4}{5}$$

$$= \frac{1}{5}$$

$$\therefore np = 5$$

$$\Rightarrow \text{Now, } 5^{23}P(X=3) = 5^{23} [{}^nC_3 p^3 q^{n-3}]$$

$$= 5^{23} \left[{}^{25}C_3 \left(\frac{1}{5} \right)^3 \left(\frac{4}{5} \right)^{22} \right]$$

$$= 5^{23} \left[\frac{25 \times 24 \times 23}{6} \times \frac{1}{5^3} \times \frac{4^{22}}{5^{22}} \right]$$

$$= 4 \times 23 \times 4^{22}$$

$$\Rightarrow 5^{23}P(X=3) = 23 \times 4^{23} \quad \dots(\text{ii})$$

From equation (i) and (ii):

$$\lambda = 23$$

103. Option (b) is correct.

x	y	xy	x^2
-4	1	-4	16
-1	2	-2	1
2	7	14	4
3	1	3	9

$$\Sigma x = -4 - 1 + 2 + 3 = 0$$

$$\Sigma y = 1 + 2 + 7 + 1 = 11$$

$$\Sigma xy = -4 - 2 + 14 + 3 = 11$$

$$\Sigma x^2 = 16 + 1 + 4 + 9 = 30$$

$$\bar{x} = \frac{\Sigma x}{4} = \frac{0}{4} = 0$$

$$\bar{y} = \frac{\Sigma y}{4} = \frac{11}{4}$$

$$b_{yx} = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{4}}{\Sigma x^2 - \frac{(\Sigma x)^2}{4}}$$

$$= \frac{11 - 0}{30 - 0} = \frac{11}{30}$$

Equation of regression line of y on x will be.

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$\Rightarrow y - \frac{11}{4} = \frac{11}{30}(x - 0)$$

$$\Rightarrow y = \frac{11}{30}x + \frac{11}{4}$$

$$\therefore a = \frac{11}{4} \text{ and } b = \frac{11}{30}$$

$$2a + 15b = 2 \times \frac{11}{4} + 15 \times \frac{11}{30}$$

$$= 11$$

104. Option (a) is correct.

Slope of line $x + 2y + 1 = 0$

$$\Rightarrow m_1 = -\frac{1}{2}$$